

Name: _____





BOOTSTRAP

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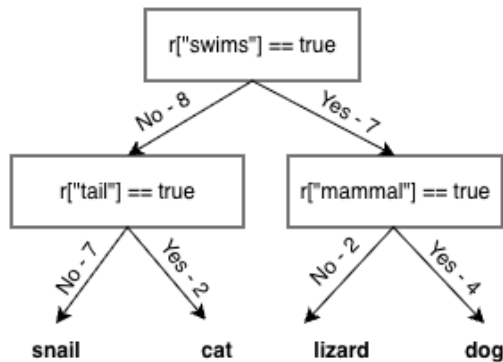
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Making Predictions with AI

One of the biggest jobs AI does is **prediction**: given an input, predict an output. Prediction comes in two flavors, depending on what kind of question we are trying to answer.

Classification ("Which One?")

A **classifier** predicts which **category** an input belongs to ("Is animal this a dog, a cat, or a snail?"). This unit covers **decision trees**, a common tool for classification.

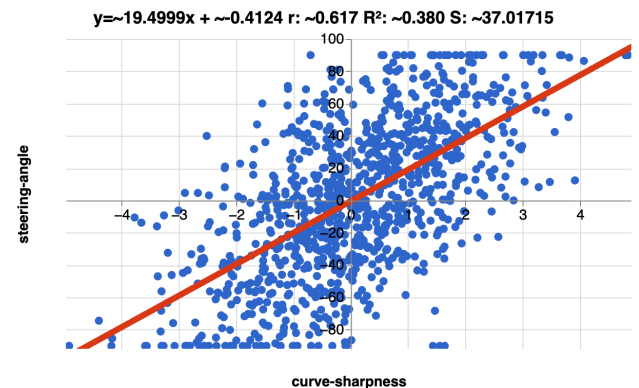


- These trees are composed of **decision nodes**, each of which asks a question to narrow down the possible category.
- Each question narrows the possibilities until only one prediction remains (called a **leaf node**)
- **Some questions are better than others.** A good split divides the data into the *neatest, most organized piles* — ideally each pile is all one category.
- A **confusion matrix** shows the accuracy of a classifier *per category* — so we can see exactly *which* mistakes a model makes, not just an overall score.

(If you've ever played "I Spy", you've built a Decision Tree yourself!)

Regression ("How Much?")

Regression predicts a **number** instead of a category ("How many degrees to turn the steering wheel?" +). The result of regression is a **function** that maps inputs to outputs.



- With **one** explanatory variable, the model is a **line of best fit** in 2D.
- With **two** variables, that line becomes a **plane of best fit** in 3D, with a **slope** for each input variable. (This gets more exotic in higher dimensions, but the principles are the same!)
- We compare how closely different models fit the data using statistical measures of fit like *S-value* .
- The math we already know maps directly onto AI vocabulary: a **variable** is a **parameter**, and a **slope** is a **weight**. Multiple regression with many weighted inputs is the engine behind a lot of real AI systems.

Supervised Learning

Whether we're predicting a category or a value, these models are built from **training data**.

- **Supervised learning** is when humans create a corpus of labeled examples — known inputs with known answers (think "answer key") — and the algorithm tries to find the pattern in these examples in order to make predictions about *new* inputs it has never seen.
- **Training** is the computation that extracts those patterns, transforming a corpus into a model. This is typically a **lossy** process: we aggregate the data and lose some of the original detail along the way. The model is a *summary* of the dataset, not a copy.

Every model in this unit — decision tree or regression — is built the same three-step way:

- (1) **Record** a corpus of human-labeled examples: known inputs paired with known answers.
- (2) **Learn** a model that captures the pattern in those examples. (This step is the "training".)
- (3) **Apply** the model to brand-new inputs it has never seen, to predict their answers.

A model that fits its training data perfectly isn't always better!

Over-fitting happens when a model "memorizes" the noise in its training data — often from highly variable quantitative columns — and then performs badly on new data it hasn't seen. The goal isn't to memorize the past; it's to **predict the future**.

What Does the Code Do?

For this page, you'll need to open the [Decision Tree Starter File](#) on your computer and click "Run".

Defining Rows

Focus on lines 16-20 in the Definitions Area.

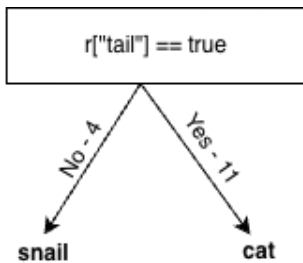
- 1) What do you think `firepaw = row-n(training, 0)` does? _____
- 2) Type `firepaw` into the Interactions area. Why are some of the cells darker than others? _____

- 3) On line 20 of the Definitions Area, define `frisky` to be the sixth row of the table. Then click "Run" again to load the code.
- 4) Evaluate `firepaw["swims"]`. What did you get? _____ Where did that come from? _____
- 5) What code would tell me if `frisky` is a mammal? _____. Test it out.

Defining our tail-classifier Function

Focus on lines 23-31 in the Definitions Area.

Read the code for `tail-classifier` carefully. Use the **decision tree** diagram to guide you.



```
# tail-classifier :: Row -> String
# consumes an animal, and predicts the species
fun tail-classifier(r):
  if r["tail"] == true:
    "cat"
  else:
    "snail"
  end
end
```

- 6) What will the function return if the animal has a tail? _____ How do you know? _____

- 7) What will the function return if the animal does NOT have a tail?? _____ How do you know? _____

Some helpful Functions

Focus on lines 37-59 in the Definitions Area.

- 8) What does `animal-img(firepaw)` evaluate to? _____ `animal-img(frisky)`? _____
- 9) In the Interactions Area, evaluate `image-dot-plot(training, "tail", animal-img)`. What happens? _____

- 10) Type `classify(training, "species", tail-classifier)` in the Interactions Area. What happens? _____

★ Make sure you know how to use `image-dot-plot` to see the distribution of the animals based on `sex`, `mammal`, `tail`, and `swims`.

Building a Decision Tree: The Algorithm

Split on	"no" leaf (most common in this group)	"yes" leaf (most common in this group)	# Correct
<code>r["tail"] == true</code>	"snail" or "lizard" or "tarantula" (2 out of 6)	"cat" (5 out of 11)	7

Use this table as a model for completing the Categorical and Quantitative Splits sections below. Notice:

- the 1st column is the code that will ask our question: "Does the animal have a tail?"
- the 2nd and 3rd columns show predicted species for animals with and without tails (and the numbers these predictions were based on)
- the last column tell us how many of these predictions would be correct for the animals in our testing table

Categorical Splits

1) Complete the table below, using `image-dot-plot` to see the distribution of `mammal`, `tail`, `swims` and `sex`.

Split on	"no" leaf (most common)	"yes" leaf (most common)	# Correct
<code>r["mammal"] == true</code>	"snail" or "tarantula" (2 of 4)	"cat" (5 of 9)	7
<code>r["tail"] == true</code>			
<code>r["swims"] == true</code>			
<code>r["sex"] == "hermaphrodite"</code>			
<code>r["sex"] == "female"</code>			

2) Why wouldn't it make sense to have a second row for `r["mammal"] == false`? _____

Quantitative Splits

3) Complete the table below, using `image-dot-plot` to see the distribution of `pounds`.

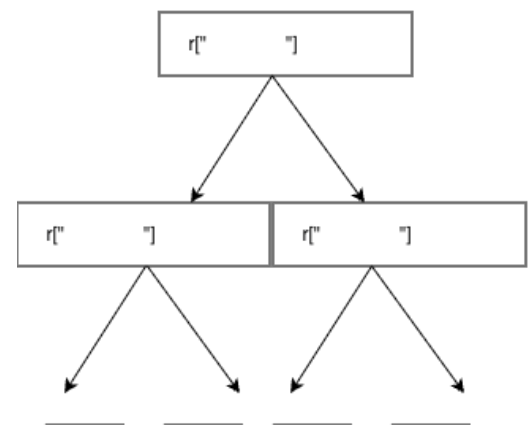
Split on	"no" leaf	"yes" leaf	# Correct
<code>r["pounds"] > 2</code>	"snail" or "tarantula" (2 of 4)	"cat" (5 of 11)	7
<code>r["pounds"] > 6</code>			
<code>r["pounds"] > 9.5</code>			

Completing the Tree

4) Choose your split and explain your choice: _____

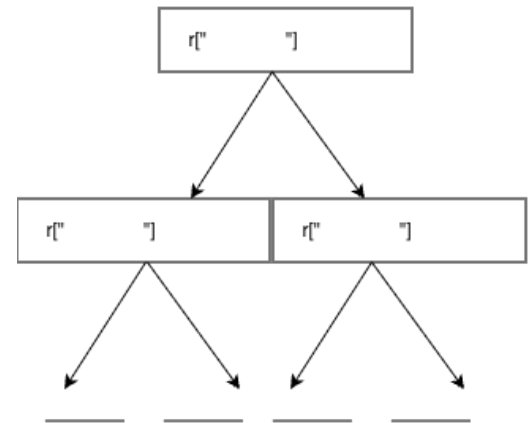
5) Fill in the root at the top of the tree below, and label the edges with the number of animals in each sub-group. Place a check next to each row of the table that your split predicts correctly.

name	species	sex	pounds	tail	mammal	swims
"Firepaw"	"cat"	"male"	9.5	TRUE	TRUE	FALSE
"Leafpool"	"cat"	"female"	10.6	TRUE	TRUE	FALSE
"Shadestar"	"cat"	"female"	9.0	TRUE	TRUE	TRUE
"Bramble"	"cat"	"female"	4.1	TRUE	TRUE	FALSE
"Brick"	"cat"	"male"	8.5	TRUE	TRUE	FALSE
"Frisky"	"dog"	"female"	9.6	TRUE	TRUE	TRUE
"Socks"	"dog"	"female"	10.0	TRUE	TRUE	TRUE
"stripes"	"dog"	"male"	11.1	TRUE	TRUE	TRUE
"Snickers"	"dog"	"male"	10.3	TRUE	TRUE	TRUE
"Bruni"	"lizard"	"female"	8.2	TRUE	FALSE	TRUE
"Jesús"	"lizard"	"male"	3.5	TRUE	FALSE	FALSE
"Gary"	"snail"	"hermaphrodite"	0.02	FALSE	FALSE	FALSE
"Turbo"	"snail"	"hermaphrodite"	0.03	FALSE	FALSE	FALSE
"Charlotte"	"tarantula"	"male"	0.35	FALSE	FALSE	FALSE
"Webs"	"tarantula"	"female"	0.4	FALSE	FALSE	FALSE



Building a Decision Tree: The Algorithm (Continued)

name	species	sex	pounds	tail	mammal	swims
"Firepaw"	"cat"	"male"	9.5	TRUE	TRUE	FALSE
"Leafpool"	"cat"	"female"	10.6	TRUE	TRUE	FALSE
"Shadestar"	"cat"	"female"	9.0	TRUE	TRUE	TRUE
"Bramble"	"cat"	"female"	4.1	TRUE	TRUE	FALSE
"Brick"	"cat"	"male"	8.5	TRUE	TRUE	FALSE
"Frisky"	"dog"	"female"	9.6	TRUE	TRUE	TRUE
"Socks"	"dog"	"female"	10.0	TRUE	TRUE	TRUE
"stripes"	"dog"	"male"	11.1	TRUE	TRUE	TRUE
"Snickers"	"dog"	"male"	10.3	TRUE	TRUE	TRUE
"Bruni"	"lizard"	"female"	8.2	TRUE	FALSE	TRUE
"Jesús"	"lizard"	"male"	3.5	TRUE	FALSE	FALSE
"Gary"	"snail"	"hermaphrodite"	0.02	FALSE	FALSE	FALSE
"Turbo"	"snail"	"hermaphrodite"	0.03	FALSE	FALSE	FALSE
"Charlotte"	"tarantula"	"male"	0.35	FALSE	FALSE	FALSE
"Webs"	"tarantula"	"female"	0.4	FALSE	FALSE	FALSE



1) In the diagram above, fill in the code for your root node and the number of animals in each branch from your work on [Building a Decision Tree: The Algorithm](#).

2) In the [Decision Tree Starter File](#), change the `my-classifier1` function to use your root. Click "Run", and evaluate `classify(training, "species", my-classifier1)`. How many did your classifier get right?

Dogs: ____ of 4 Cats: ____ of 5 Lizards: ____ of 2 Snails: ____ of 2 Tarantulas: ____ of 2

TOTAL: ____ of 15

Choosing a Split for Animals on the Left

What is the best split to use for the animals on the left?

Split on	"no" leaf (most common)	"yes" leaf (most common)	# Correct
<code>r["mammal"] == true</code>	"snail" or "tarantula" (2 of 5)	"cat" (4 of 4)	6
<code>r["tail"] == true</code>			
<code>r["sex"] == "hermaphrodite"</code>			
<code>r["sex"] == "female"</code>			

Choosing a Split for Animals on the Right

What is the best split to use for the animals on the right?

Split on	"no" leaf (most common)	"yes" leaf (most common)	# Correct
<code>r["mammal"] == true</code>	"lizard" (1 of 1)	"dog" (4 of 5)	5
<code>r["tail"] == true</code>			
<code>r["sex"] == "hermaphrodite"</code>			
<code>r["sex"] == "female"</code>			

3) Choose your splits, filling in the second level of the decision tree diagram at the top of this page.

4) Complete the `my-classifier2` in the starter file. Click "Run", and evaluate `classify(training, "species", my-classifier2)`. How many did your classifier get right?

Dogs: ____ of 4 Cats: ____ of 5 Lizards: ____ of 2 Snails: ____ of 2 Tarantulas: ____ of 2

TOTAL: ____ of 15

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